

An Interactive Workplace for Improving Human Robot Collaboration

Sketch Workpiece Interface for Fast Teaching (SWIFT).

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ABSTRACT

Ubiquitous computing refers to the concept of integrating computers and technology into everyday objects and environments, making them constantly available and seamlessly interconnected. In recent times, there has been a growing adoption of the concept in industrial settings as well. This contribution targets the replacement of an industrial manual repair process through a robotic solution. The focus of this work is on the novel interface utilizing a digital pen and spatial augmented reality, which mimics the original job preparation process but ubiquitously enables automated robot programming. This shifts the work of the human from a dull, dirty and dangerous process to the cognitively more demanding part of inspection and process strategy definition, whereas the robot is used as a dexterous and tenacious intelligent process tool to perform the surface processing operation itself.

CCS CONCEPTS

• **Computer systems organization**; • **Embedded and cyber-physical systems**; • **Robotics**; • **External interfaces for robotics**; • **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Mixed / augmented reality**; • **Ubiquitous and mobile computing**; • **Ubiquitous and mobile computing theory, concepts and paradigms**; • **Ambient intelligence**;

KEYWORDS

Industrial Extended Reality, Human Robot Interaction, mixed reality robot programming, pen-based interaction

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1 INTRODUCTION

The material for mineral cast products (e.g. bathtubs or washing basins) belongs to the class of composite materials. In order to give the material sufficient strength, acrylic polymer consisting of various curing agents and resin is mixed with natural mineral substances. Mineral castings are manufactured in five steps.

Step 1 regards the manufacturing and preparation of molds. Molds are produced as assemblies of mold parts to be able to demold products by dismantling the casting mold. Before casting, the insides of the casting mold segments must be cleaned and polished in order to get rid of residuals from prior castings. This is usually done in manual labor. To prevent the mineral casting from sticking to the inside of the mold segments, the entire inside surface of the mold is coated with a special release agent, usually by spraying. Step 2 refers to dosing and mixing the individual material components before the liquid mineral is poured (step 3) into the mold. Air bubbles that form during the casting process have to be knocked out before the drying phase (step 4). The binder hardens under only slight heating. The surfaces usually often already meet the requirements after demolding and washing off the release agent. Occasionally burrs appear e.g. at the joining surfaces of the casting mold segments, which have to be repaired (step 5) together with other surface defects such as breakouts, scratches or surface shrinkage. These can be removed by sanding off with hand grinders, filling, sanding and polishing. Repair and rework of casting molds (during step 1) as well as repair of imperfections in the final product during step 5 are time consuming, unergonomic and performed by manual labor and for obvious reasons meet the requirements: dull, dirty and dangerous (otherwise known as the 3 Ds of robotization) that characterize tasks predestined for automation.

Surface finishing processes are characterized as challenging tasks since each defect requires an individual repair strategy. In general, available industrial surface repair solutions mostly target serial production processes [1]. In contrast to this, mineral casting products are produced in high-mix low-volume (HMLV) production programs, which typically require changeover, setup and commissioning more frequently than in series production. These requirements

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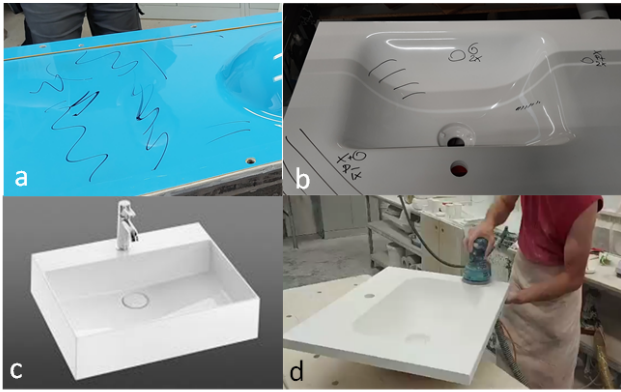


Figure 1: Mineral Casting Production. (a) mold with markings, (b) product with markings, (c) product (www.helopal.at), (d) manual processing.

make automation economically unviable in many other cases relevant for small and medium enterprises that manufacture products in artisanal fashion [2].

Figure 1 shows mineral casting products as well as intermediate steps during manufacturing. Workers in preparation of manual processing of molds and products mark defective regions on molds and workpiece with waterproof fiber markers in order to increase visibility. These marks vanish during the processing with abrasive sanding and polishing equipment.

Regarding the robotic automatization, the usage of a Collaborative Robot (cobot) seems to be suitable. Cobots are designed to be safe, flexible, and easy to use, allowing for seamless integration into existing workflows and processes. By combining the strengths of human operators and robots, this can improve efficiency, accuracy, and quality while reducing costs and minimizing waste.

Human operators can perform tasks that require judgment, dexterity, and decision-making, while cobots can perform repetitive or time-consuming tasks such as surface processing. Human robot interaction (HRI) focuses on designing and developing robots that can effectively interact with humans in various settings. It involves understanding human behavior and preferences, as well as the capabilities and limitations of robots. Human-centered automation (HCA) is an approach to designing and implementing automated systems that prioritize the needs, abilities, and preferences of human operators. The goal of HCA is to create automation that works in partnership with humans, rather than replacing them. In HCA, the design of automated systems considers the human factors involved, including cognitive and physical capabilities, task requirements, and work environment. This approach aims to make automation more intuitive and user-friendly, which can lead to better performance, greater efficiency, and reduced error rates. HCA involves a user-centered design process that incorporates feedback from human operators and other stakeholders. Thus, the combination of concepts from HCA, HRC and HRI can enhance the overall quality of work by reducing physical and cognitive stress on human operators, allowing them to focus on more complex, rewarding tasks or tasks that currently cannot be automated. It can also help to

improve worker satisfaction and retention, as workers are able to work in safer and more engaging environments.

2 RELATED WORK

2.1 Human Robot Interaction

Human Computer Interaction (HCI) in general, and Human Robot Interaction (HRI) in particular requires ergonomic interface modalities that consider the ways in which people interact with the physical world. Yohannan et al. [13] discuss touch interaction with social robots. Berg et al. [9] give an overview of literature on interfaces utilized in an industrial robot and service robot context that also includes modalities like touch interaction, gestures and voice control providing specific advantages and disadvantages.

Huang et al. [14] illustrates the great potential of direct mapping of interaction between digital interfaces and physical affordance by converging wearable devices and augmented reality (AR) technology. The great number of modalities and approaches for bidirectional communication opens a multidimensional room of combination possibilities that needs to be filled in an application specific way that allows users to utilize industrial robots in an ergonomic, preferred and productive fashion.

2.2 Pen-Based Interaction in HRI

Pen-based interaction has proven to be promising in HRI and several studies investigated the usability of their applications. In ubiquitous computing already [15] envisioned a pen-based interaction with 3D models, a concept picked up lately by the robotics community. So-called ‘instrumented tools’ were proposed for robot programming in automotive pre-assembly processes [16]. In [17], a digital pen was developed that was used to program robots to support welding tasks by hand-writing on objects. The authors of [18] compared different interaction modalities for defining accurate positions and found 3D pens as the most preferred input modality in comparison to touch or speech interaction. The evidence presented here suggests pen-based interaction as an interesting technological candidate for surface repair processes.

2.3 Extended Reality on industrial workplaces

Various studies have assessed the efficacy of Extended Reality systems and applications for industrial settings, see [19, 20]. For example, [2] introduced a novel solution that projects surface metrology data directly on a polished component by means of a head mounted display (HMD) worn by an operator to assess the quality of the surfaces. However, [21] revealed that while the projector-mirror setup shows clear improvements in safety and work ergonomics, HoloLens-based Augmented Reality (AR) is not yet suitable for industrial manufacturing. To overcome technological and ergonomic limitations of conventional AR systems, projection devices [22] have been investigated in various studies. For example, [23] used conventional projector and camera equipment to display safety areas and detect safety violations for physical HRC. The authors of [24] found higher user effectiveness and satisfaction with the information projected directly on the physical object in comparison to monitors or printed text. Similarly, [25] demonstrated that participants experienced higher satisfaction and completed the task faster when instructions were projected onto objects rather than

printed on a paper or displayed using a mobile device. Altogether, projection on work pieces is a promising method for collaborative HRI.

2.4 Automating spot repair, polishing and similar tasks

Currently in manufacturing, the assessment and surface finishing of complex parts is often done manually by skilled workers. A number of studies have begun to examine different approaches to partially automate surface finishing processes, see [3-5]. While robots are consistent and tireless, they are slower and inflexible compared to skilled workers, see [6]. Current computer vision technologies - utilized for defect detection - are yet limited in precision and robustness [7]. Thus, a collaborative approach is suitable in which a skilled worker guides and supervises a robot. However, this complementary approach still faces some HRI challenges. While traditional robot programming methods (Learning-by-programming) require expertise in coding and a significant time investment [8], advanced programming or robotic learning paradigms (e.g., Learning-by-interaction, Learning-by-demonstration) requires novel interfaces to facilitate intuitive interaction with the robot [9]. Additionally, recent studies have discussed how intuitive user interfaces for robot programming paradigms rely also on multi-modal interaction and extended reality solutions, see [10-12]. Hence, collaborative robots in the context of spot repair require intuitive interfaces to support skilled workers.

3 METHOD

3.1 Human Centered Automation Design

A human-centered design process, considering the process introduced in Section 1, identified the time-consuming surface processing operations as most suitable for processed by the robot, whereas the identification and classification of surface imperfections as well as the definition of processing strategies were considered preferably as human tasks since closing the loop with advanced surface inspection systems as well as control systems were considered technically or economically unviable.

In designing digital user interfaces that utilize analog inputs, HCI practitioners consider the ways in which people interact with the physical world through senses such as touch, visual cues, and sound. They also consider the various analog interfaces that people may be familiar with. The most intuitive interface is an interface that is not perceived as a digital interface. Utilizing the markings of the waterproof fiber marker a human artisan would create for his own job preparation. The identification and analysis of markings on objects with pronounced 3D curvature or surface attributes challenging for visual inspection (e.g. highly polished reflective surfaces) is difficult. Ubiquitous computing refers to the concept of integrating computers and technology into everyday objects and environments, making them constantly available and seamlessly interconnected. Therefore, the integration of available technology in terms of available digital input devices (3D pens – see Section 2) with projection systems for immediate feedback in order to create a digital version of analog input devices is obvious. [26] identified the described pen-based interaction as preferable to screen-based

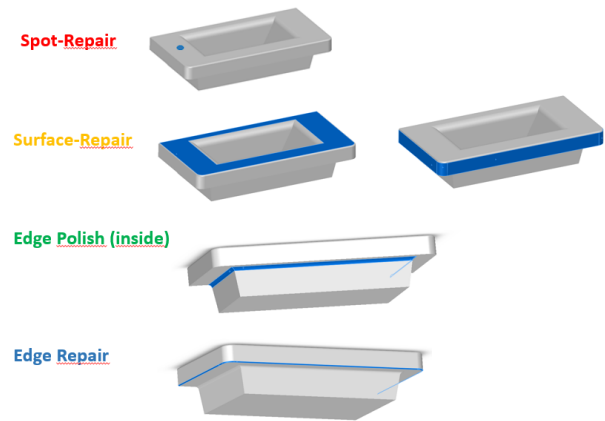


Figure 2: surface processing operation types.

interaction. The implemented UI technology is utilized for the parameterization of two specific surface repair operation types – spot and surface repair (see Figure 2) – which can be distinguished with respect to the physical size of the processing area. Edge polish and repair operations are currently not considered.

3.2 Demonstration Video

The demonstration video showcases the application of the implemented user interface for the two specific surface repair operation types – spot and surface repair, see Figure 2. First the equipment utilized is introduced. A 3D sensor articulated together with a cobot on a linear axis is scanning the workspace acquiring point clouds. Object detection algorithms identify the location of CAD objects within the point clouds. The result of the detection is projected back to the workspace indicating a successful identification of the position of the workpiece to the worker. A 3D tracking device is utilized for the implementation of the digital pen device. For spot repair operations the digital pen is utilized to mark spots (as relatively small and limited areas) to be processed that are filled with meandering patterns with preceding identification of a boundary polygon shape. For surface repair operation marking entire surfaces would foil the attempt to shorten parameterization time with digital pens. Therefore, surface segmentation with subsequent selection of a specific subsurface followed by process planning takes place. In order to give the operator feedback on the toolpaths planned they are projected on the surface. In similar fashion areas processed by the tool are highlighted. Figure 3 shows the setup discussed.

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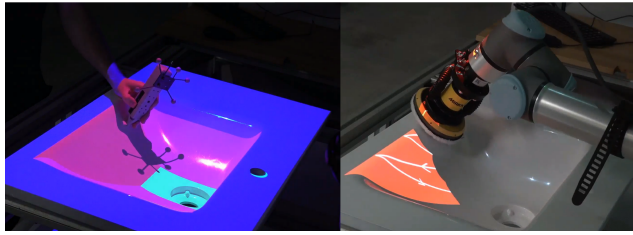


Figure 3: CoPolishing Process, (left) pen based surface selection, (right) cobot enabled processing.

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